# TAMID QUANT — STRATEGY ONE PAGER

## Bitcoin Digital Options: Model-Prior Tail Model (#5)

Date: November 2026 | Presenter: Marcos Garay

## The Idea

*Kalshi's hourly "Bitcoin above $K at 5pm ET" contracts overprice longshots in the last hours before expiry. The tail strategy sells overpriced low-tail YES and buys underpriced high-tail YES, sizing each day so the worst BTC settlement loses at most 3% of bankroll. Model #5 changes one thing: its estimate of each contract's chance of YES starts from a volatility model's probability instead of the market price, then learns from past outcomes. Over all the data (Dec 2024 to Nov 2026) it turned $1,000 into $1,129.92, against $1,052.61 for the base, with a Sharpe ratio of 1.17 against 0.47 and the smallest drawdown of any model (−$31.54). It made money in every period, but the gain is not yet statistically significant.*

## Mathematical Framework

**Core Model:** *Shrink past outcomes toward the volatility model, per contract:*

$f = (\Sigma y + N_0 \cdot p_{model}) / (n + N_0)$, $N_0 = 50$, where n and Σy are the prior count and YES count in the contract's (horizon, price range), and $p_{model}$ is the contract's own model probability. The base used the range's average market price in place of $p_{model}$.

**Why it helps:** *The model prices tails closer to reality than the market:*

contracts priced 2-5c had a model probability of 1.5% and settled YES 0.8%. Example: a 3c contract in a range with 150 prior quotes and 1 YES gets f = 1.25% in the base and 0.88% in #5 (model at 1.5%), so its edge is larger. The model is an EWMA volatility forecast with filtered-historical-simulation tails.

**Entry / Exit Logic:** *Same as the base: decide 6h and 3h before expiry, fill one hour later:*

Trade mid < 10c or > 90c, spread ≤ 6c, edge ≥ 0.5c (sizing price = quote ± 2c, fee $0.07 \cdot P \cdot (1 - P)$), at least 100 prior observations. Hold to settlement.

**Position Sizing:** *Same linear program as the base:*

maximize $\Sigma\, q_i \cdot edge_i$ so that every BTC settlement scenario loses at most 3% of bankroll, $0 \le q_i \le 50$.

**Selection:** *One of ten numbered models, tuned Jan to Jun 2026, scored once on Jul 1 to Nov 11:*

#5 and #7 beat the base in both periods. #5 has no setting to tune.

## Key Risks

- Not significant yet: Sharpe minus base over all data = +0.69 (95% CI −0.61 to +2.17); mean daily return +0.019% (95% CI −0.005% to +0.040%)
- Tail days remain: 7 days lost about $30. The average winning trade makes +$0.65 and a loser −$28.39, so one tail day erases about 44 winning trades
- It relies on the volatility model being right in the tails. A sudden jump the model underprices hits exactly the contracts #5 trades most
- On 31 days the realized worst case ended past the 3% cap (worst 3.17%), because fills come an hour after sizing
- Most of the profit ($89 of $130) came from buying high-tail YES 6h out

## At a Glance

| | |
|---|---|
| Asset Class: | *Crypto / Prediction Markets* |
| Strategy Type: | *Longshot Bias / Model-Anchored Estimate* |
| Direction: | *Short Low-Tail YES, Long High-Tail YES* |
| Holding Period: | *3-6 Hours (Hold to Expiry)* |
| Universe: | *Kalshi KXBTCD Tail Strikes* |

## Backtest Results

| Period | #5 Sharpe | Base Sharpe | #5 profit |
|---|---|---|---|
| Before 2026 | 0.71 | n/a | +$22 |
| Tuning | 1.26 | 0.29 | +$45 |
| Holdout | 1.88 | 1.07 | +$63 |
| All data | 1.17 | 0.47 | +$130 |

*Tuning Jan-Jun 2026, holdout Jul 1-Nov 11 2026, all data Dec 23 2024-Nov 11 2026. Annualized Sharpe ratio; base made 2 trades before 2026. 523 trades on 332 of 688 days; max drawdown −$31.54; never below $1,000. $1,000 bankroll, 3% worst-case cap per day.*

## Data Requirements

- Kalshi KXBTCD hourly bid, ask and settlement; Coinbase BTC-USD hourly candles - public APIs
- 815 expiries (Jul 2024 to Nov 2026), 36,093 quotes at 6h and 3h; the first months only train the estimate

## Key Parameters

- Shrinkage $N_0 = 50$ toward the model probability
- Worst-case cap 3% of bankroll per expiry day
- Tails mid < 10c or > 90c at 6h and 3h; min edge 0.5c; slippage 2c; max 50 contracts per order

## Open Questions

- Does it hold on data after Nov 11, 2026 (a true second holdout)?
- It traded profitably in 2025 when the base could not: is that edge real or luck (84 trades)?
- Would a stronger or weaker pull toward the model ($N_0$) do better?
- Is there depth to trade 30+ contracts at these prices?